

Astha Tripathi

B.Tech, Computer Science and Engineering | CGPA: 8.86
JSS Academy Of Technical Education, Noida | 2020–2024

Tel: +91-6387230296
Email: asthatripathics174@gmail.com
GitHub github.com/albetros24
LinkedIn linkedin.com/in/astha-tripathi

SUMMARY

Application SRE and Platform Engineer with 2+ years at Zeta (Directi) — AWS, Kubernetes, Terraform, and full-stack observability (Prometheus, Grafana, OpenSearch/ELK). Proven track record in zero-downtime EKS operations, high-severity incident response, and Go/Python automation to eliminate operational toil.

TECHNICAL SKILLS

Languages & Scripting: Go (Golang), Java, Python, Shell scripting, SQL
Cloud & IaC: AWS (EKS, EC2, S3, Load Balancer, EBS), Terraform, Helm, Serverless Framework
Containers & Orchestration: Kubernetes (Operators, CRDs, Prometheus-Operator), Docker
Observability & Logging: Prometheus, Grafana, Alertmanager, Node Exporter, OpenSearch, Elasticsearch (ELK-compatible)
Backend & Data: Microservices, RESTful APIs, MongoDB, Redis, CI/CD pipelines

EXPERIENCE

- **Zeta (Directi)** July 2025 – Present
Software Engineer 1 – Platform & Backend (Application SRE) Bengaluru, India
 - Orchestrated a **zero-downtime EKS upgrade from v1.32 to v1.35** across all production clusters, maintaining 100% service availability; authored runbooks and post-upgrade validation scripts in **Python and Shell**.
 - Managed full-lifecycle migration of **AWS Load Balancer Controller, EBS CSI drivers, and Node Exporter** across multi-region environments using **Terraform** and **Helm**, with zero production incidents.
 - Migrated search infrastructure from **Elasticsearch to OpenSearch** (ELK-compatible), designing custom dynamic mappings that reduced indexing time by ~30%; architected a **Redis cache-backed search layer** cutting p95 latency from ~450ms to ~120ms.
 - Drove reliability and fault-tolerance improvements for **Pluxee App** (India & Philippines), optimizing high-traffic API paths serving millions of daily transactions via circuit-breaker patterns and graceful degradation.
- **Zeta (Directi)** July 2024 – July 2025
Software Engineer – SRE & Backend Bengaluru, India
 - Designed and deployed a **full observability stack** (Prometheus, Grafana, Alertmanager) across production Kubernetes clusters; reduced MTTD for critical incidents and enabled proactive capacity planning.
 - Led incident response and root cause analysis for high-severity production issues; conducted post-incident reviews and drove long-term corrective actions across microservices to prevent recurrence.
 - Improved **AWS security posture from 69% to 76%** via Python-automated configuration remediation and secure coding enforcement across 10+ services, meeting cloud compliance standards.
 - Reduced **Pluxee App API latency by ~25%** through profiling, SQL query optimization, and connection pool tuning; deployed fixes via **CI/CD pipelines** with automated rollback on failure.
- **Zeta (Directi)** Feb 2024 – July 2024
Software Engineer Intern – SRE Bengaluru, India
 - Built a **Go-based Kubernetes Operator** to automate deployment health monitoring and real-time alerting via Microsoft Teams — eliminating manual toil and reducing on-call intervention.
 - Configured **Prometheus-Operator** for infrastructure-wide metrics; wrote **Python and Shell automation scripts** to streamline deployment pipelines via Serverless Framework across staging and production.

PERSONAL PROJECTS

- **RedeployBot**
Go, Kubernetes, Operator SDK, Helm, Docker, Terraform
 - Engineered a **custom Kubernetes CRD Operator in Go** that automates deployment lifecycle management; performs pre-restart health assessments via Go goroutines, blocking unsafe restarts and **eliminating manual operational toil**.
 - Integrated **Microsoft Teams** alerting for unhealthy pods and post-deployment stability checks — mirroring production SRE incident response and on-call notification workflows.